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**Dental Anomaly Detection through Biomedical X-rays: Deep
Learning Approach**

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Dental Anomaly Detection through Biomedical X-rays: Deep Learning Approach

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SUMMARY

This study explores the design and implementation of an advanced AI-driven system for identifying dental anomalies through the application of biomedical imaging to revolutionize diagnostic processes.

The central interest is to use dental panoramic X-rays as input for a deep learning-based approach, using the YOLOv8 model (You Only Look Once version 8), which is widely recognized as efficient and very accurate for object detection tasks.

An artificial intelligence algorithm detects together with organizes different dental issues which include root canals as well as crowns and fillings and extractions and impacted teeth and implants. The technology revolutionizes dental diagnostics since it produces precise outcomes and needs fewer operators and enhances dentist assurance.

The research project seeks to enhance dental treatment delivery by overcoming existing boundaries which affect conventional diagnostic approaches. The situation becomes crucial in regions which lack sufficient medical professional availability. The present method requires extensive time along with advanced expertise which creates delays in both diagnosis and subsequent treatment procedures.

The system proposed here automates the detection process, which speeds up diagnostics, reduces the chance of human error, and ultimately improves the quality of dental care.

It is particularly valuable in areas with limited access to experienced dentists because the system operates with minimal user input, offering high-quality diagnostic capability even in low-resource settings.

Image processing combined with deep learning algorithms produces faster diagnosis speed while simultaneously improving dental evaluation precision.

The technology assists dentists with clinical decision-making through a dependable solution that fits seamlessly within their current practice patterns. The system fills an enduring dental informatics gap with its scalable precision and accurate detection of dental abnormalities.

It can identify numerous various dental conditions and classify them into distinctive anomaly classes, a critical intermediate step in diagnosis and treatment planning.

This study is one illustration of how the future holds with AI potentially changing the practice of dentistry for improved precision and effectiveness in dental checks and diagnosis, thereby likely leading to significantly improved outcomes in patients through AI solutions.

The delivery of health services could experience improvement as technology progresses which opens up possibilities for further advancements.

The project establishes dental informatics innovation potential while proving technology benefits patient care and simplifies clinical workflow.

ABSTRACT

This paper suggests a novel approach to dental anomaly detection by leveraging the strength of deep learning models, and more precisely, the YOLOv8 architecture, to address classification issues on dental abnormalities based on panoramic X-ray images.

A customized dataset of 800 high-resolution images was meticulously developed to train the model, and it achieved an impressive mean Average Precision (mAP50) of 95.3%.

The study shows how the model offers substantial diagnostic improvements together with increased efficiency for clinical dental practice. The system combines sophisticated image processing approaches with deep learning algorithms to speed up diagnostics while improving dental assessment accuracy.

The system functions to support dentists by delivering an effective diagnostic tool which fits smoothly into current clinical procedures. The system proves essential for quick precise diagnosis of multiple dental diseases including basic and advanced anomalies in both regular and difficult diagnostic situations.

This study addresses a seminal need in dental informatics by proposing a rapid and scalable framework for the identification of abnormal dental states. The platform demonstrates an essential capability to detect multiple dental conditions by classifying them as anomaly categories which support healthcare providers in identifying dental abnormalities properly.

This research demonstrates the broad potential of artificial intelligence-based solutions which will transform dental patient services through advanced evaluation accuracy and rapid processing. The project demonstrates how AI will transform dentistry by creating personalized treatment diagnostic plans which will become more efficient as technology evolves.

In conclusion, AI will transform dentistry into a new paradigm through this paper which explains how personalized and efficient treatment options will emerge from future advancements. The future development possibilities stand enormous because advancing technology presents vast opportunities to improve oral health care delivery.

Keywords: YOLOv8, Image Processing, Dental Informatics, Clinical Dentistry, Machine Learning, Artificial Intelligence, Neural Networks

ÖZET

Bu çalışma, panoramik diş röntgenleri üzerinden diş anomalilerinin sınıflandırılması sorununu ele almak amacıyla, derin öğrenme modellerinin ve özellikle YOLOv8 mimarisinin gücünden yararlanan yenilikçi bir yaklaşım önermektedir.

Modeli eğitmek için özenle geliştirilen, yüksek çözünürlüklü 800 görüntüden oluşan özel bir veri seti kullanılmış ve model, %95,3 mAP50 (ortalama doğruluk) başarısı elde etmiştir.

Çalışma, modelin klinik diş hekimliği uygulamalarında hem tanısal doğruluğu önemli ölçüde artırdığını hem de süreç verimliliğini geliştirdiğini göstermektedir. Sistem; gelişmiş görüntü işleme tekniklerini, derin öğrenme algoritmalarıyla birleştirerek, hem teşhis hızını artırmakta hem de diş değerlendirme doğruluğunu yükseltmektedir.

Bu sistem, diş hekimlerinin mevcut klinik süreçlerine sorunsuz bir şekilde entegre olabilen etkili bir teşhis aracı sağlamaktadır. Sistem, özellikle temel ve ileri seviyedeki çürükler de dahil olmak üzere, birden fazla diş hastalığının hızlı ve doğru şekilde teşhis edilmesinde kritik bir rol oynamaktadır.

Çalışma, diş bilişiminde önemli bir ihtiyaca cevap vererek, anormal diş durumlarının tespiti için hızlı ve ölçeklenebilir bir yapı önermektedir. Geliştirilen platform, farklı diş hastalıklarını anomali kategorileri olarak sınıflandırarak, sağlık uzmanlarının diş anomalilerini doğru bir şekilde tespit etmesine yardımcı olacak temel bir yetenek sergilemektedir.

Bu araştırma, yapay zeka tabanlı çözümlerin, ileri seviye değerlendirme doğruluğu ve hızlı işlem gücü sayesinde diş hekimliği hizmetlerini dönüştürme potansiyelini geniş bir şekilde ortaya koymaktadır. Proje, yapay zekanın kişiselleştirilmiş teşhis planları oluşturma yoluyla diş hekimliğinde nasıl bir dönüşüm yaratacağını ve teknolojinin ilerlemesiyle bu süreçlerin nasıl daha da verimli hale geleceğini göstermektedir.

Sonuç olarak, bu çalışma ile yapay zekanın diş hekimliğinde kişiselleştirilmiş ve etkili tedavi seçenekleri oluşturma yolunda nasıl yeni bir paradigma yaratacağını ortaya koymaktadır. Gelecekteki teknolojik gelişmeler, ağız ve diş sağlığı hizmetlerini daha da iyileştirmek için büyük fırsatlar sunmaktadır.

Anahtar Kelimeler: YOLOv8, Görüntü İşleme, Diş Bilişimi, Klinik Diş Hekimliği, Makine Öğrenimi, Yapay Zeka, Sinir Ağları

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SYMBOLS AND ABBREVIATIONS

AI: Artificial Intelligence

DL: Deep Learning

YOLOv8: You Only Look Once version 8 (real-time object detection model)

CLAHE: Contrast Limited Adaptive Histogram Equalization (image enhancement technique)

mAP: mean Average Precision (detection accuracy metric)

IoU: Intersection over Union (bounding box evaluation metric)

REST API: Representational State Transfer Application Programming Interface

OpenCV: Open-Source Computer Vision Library

Flask: Python web framework (backend implementation)

F1-score: Harmonic mean of precision and recall (performance metric)

Panoramic X-ray: Rotational dental radiograph capturing full mouth

1. INTRODUCTION

The latest advancements in artificial intelligence (AI) and machine learning have revealed new avenues in medical diagnostic sciences, establishing unprecedented pathways toward precision and efficiency. These technological innovations are transforming healthcare delivery across various specialties, with dentistry being no exception.

The conventional dental practice requires manual inspection of anomalies which increases the risk of human mistakes. The manual inspection process takes both time and produces inconsistent diagnostic results because it depends on practitioner skill level. This study project uses deep learning and biomedical image processing tools to create an AI system that revolutionizes dental anomaly detection.

1.1. Problem statement

The detection of dental anomalies through traditional methods consumes a significant amount of time because human operators need to perform extensive manual tasks which leads to diagnostic mistakes and delayed process completion. The standard diagnostic methods create several main obstacles:

1. The process needs a significant duration for qualified professionals to complete
2. The diagnostic methods depend heavily on the operator which generates variations in the outcome
3. These diagnostic techniques prove challenging to implement in healthcare settings where specialized professionals have limited availability
4. The diagnostic methods cause treatment plans to be postponed and treatment execution to be delayed

This study presents a novel automatic deep learning solution which accelerates dental diagnostic procedures with improved precision to lower human expertise requirements and decrease diagnostic inaccuracies.

1.2. Project Purpose

The research aims to produce a diagnostic system that will help dental professionals deliver prompt and accurate diagnostic services through panoramic dental X-ray analysis. This tool functions to achieve the following objectives:

1. The system helps improve patient treatment quality by delivering better diagnostic outcomes
2. The tool shortens medical processes through its advanced automated system
3. The system provides better quality diagnostics to dental practices located in areas lacking specialist dental professionals
4. The automated system helps practitioners by handling the identification process of standard dental anomalies Digital automation allows dental health professionals to improve healthcare quality and maintain consistent results while making clinical operations more efficient.

1.3. Project scope

The project includes data handling operations from acquisition through deployment that consist of:

1. The process of collecting data and its preparation involves building a distinct dataset from dental panoramic X-rays
2. Preprocessing images together with augmentation techniques to optimize model accuracy
3. A YOLOv8-based deep learning model receives development and training for performing anomaly detection tasks
4. The model undergoes thorough assessment with suitable evaluation measures
5. A user-friendly interface gets created for dental practice integration through comprehensive design and implementation procedures

Technical requirements for the system guarantee its robustness and scalability to meet different clinical needs while preserving adaptability to multiple settings. The methodical approach leads to a solution which meets both technical standards and practical dental practice needs.

1.4. Project Objectives and Success Criteria

This project aims to achieve the following objectives:

1. A deep learning model will be developed for dental anomaly detection and classification with high precision
2. Human diagnostic capabilities will be matched or exceeded by the system's performance indicators including precision, recall and F1-score
3. An easy-to-use system interface will be developed which dental practitioners can handle effectively with brief training
4. The system must demonstrate dental X-ray processing and analysis capabilities for real-time decision support in clinical practice
5. The system will be put through rigorous testing to determine its performance levels

Primary performance metrics such as precision and recall and F1-score will determine system success alongside its clinical practice acceptance and adoption. The primary purpose of the system is to prove its capability for enhancing dental diagnostic precision and operational efficiency in actual clinical environments.

1.5. Report Outline

This document organizes the research findings to deliver a complete presentation of the research work.

The content comprises:

1. Introduction: Describing the research problem alongside its objectives and the project boundaries and organizational framework.
2. Related Studies: Analyzing current literature and system implementations to show the necessity of the proposed solution.
3. Methodology: The description of data sources and software tools and hardware components used to build the dental anomaly detection system.
4. Experimental Results: Analysis of the system's evaluation results through performance metrics.
5. Argument: The report presents an analysis of how the proposed system surpasses conventional techniques.
6. Results: The research provides a brief summary of its major discoveries and research achievements.

The report ends with an examination of research consequences and future work recommendations. The project development and its results receive confirmation through supplementary materials which deliver complete insight into dental diagnostic system capabilities.

2. RELATED STUDIES

The dental diagnostic field has made remarkable progress as artificial intelligence and machine learning systems became integrated into their operations. This chapter summarizes relevant literature and existing systems, identifies the limitations of current methodologies, and establishes the need for more advanced solutions.

The discussion sets the stage for introducing the proposed system that addresses these limitations by utilizing the YOLOv8 model for dental anomaly detection.

2.1. Existing Systems

The detection of dental anomalies in current systems relies heavily on human analysis of X-ray images which results in subjective outcomes. The development of diagnostic imaging methods has progressed throughout the years but the interpretation process fundamentally depends on practitioner skills.

New technologies have started to implement AI for dental diagnosis but their present methods use basic image processing algorithms that fall short of state-of-the-art AI capabilities. The initial implementations of deep learning in dental diagnostics have shown promise but they have not yet reached their full potential.

A number of research projects from recent times have utilized deep learning models to execute dental condition classification tasks. The research conducted by Almalki and colleagues in 2022 through orthopantomography X-ray images demonstrated that deep learning improves diagnostic precision for dental applications. Their study achieved positive results by applying convolutional neural networks to identify dental conditions.

The work by Pereira in 2020 demonstrated how convolutional neural networks could enhance diagnostic capabilities through the classification of 3D medical images which included dental scans.

However, many of these existing systems face challenges in scalability and integration into clinical practice. Limitations in its design and implementation point to the need for more innovative and comprehensive solutions.

2.2. General Problems of Existing Systems

The process of manually interpreting dental X-rays creates various important barriers which reduce diagnostic precision and operational effectiveness:

1. Time Consumption: The traditional analysis process takes up staff hours for detailed examination which could otherwise be spent on patient treatment.
2. Human Error: The human interpretation method leads to different conclusions and

diagnostic errors because of personal judgment which particularly affects intricate and subtle cases.

3. **Limited Scalability:** The manual workflow lacks the flexibility needed to expand its operations for additional patients or to help communities without proper dental care.

4. **Accessibility Barriers:** The need for expert interpretation creates access limitations especially for areas that lack qualified dental specialists.

Automated diagnostic systems need to be developed urgently because they will boost diagnostic precision while reducing the diagnostic workload on dental practitioners. The YOLOv8-based system offers a better solution for dental anomaly detection because it solves these challenges through improved efficiency and accessibility.

2.3. Comparison Between Current and Proposed Method

YOLOv8-based system integration developed within this study represents a meaningful innovation when compared to traditional operations because of its speed, performance level and user-friendly features.

YOLO (You Only Look Once) models are recognized for their exceptional balance of accuracy and computational efficiency. They are remarkably compact and can be deployed on both high-end and resource-constrained systems. This versatility creates opportunities for deployment across diverse environments, including cloud-based frameworks and local clinical settings.

The implementation of deep learning technology allows systems to automatically identify irregularities with precise results which eliminates most manual tasks and speeds up diagnostic procedures.

Table 1: Comparison of Methods

Feature	Manual Method	Proposed Method
Interpretation type	Manual interpretation	Automated interpretation
Accuracy	Inconsistent accuracy	Fast, accurate, user-friendly
Scalability	Low scalability	Scalable, innovative
Accessibility	Requires specialist expertise	Minimal expertise required

Table 1 provides a comparative analysis of traditional manual methods versus the proposed YOLOv8-based approach.

The YOLOv8 model architecture consists of two primary components: The backbone, and the head, both with a fully convolutional architecture model employed. The

YOLOv8 architecture effectively processes the input image, producing the appropriate output and improving the models machine learning capabilities to identify and detect anomalies with precision.

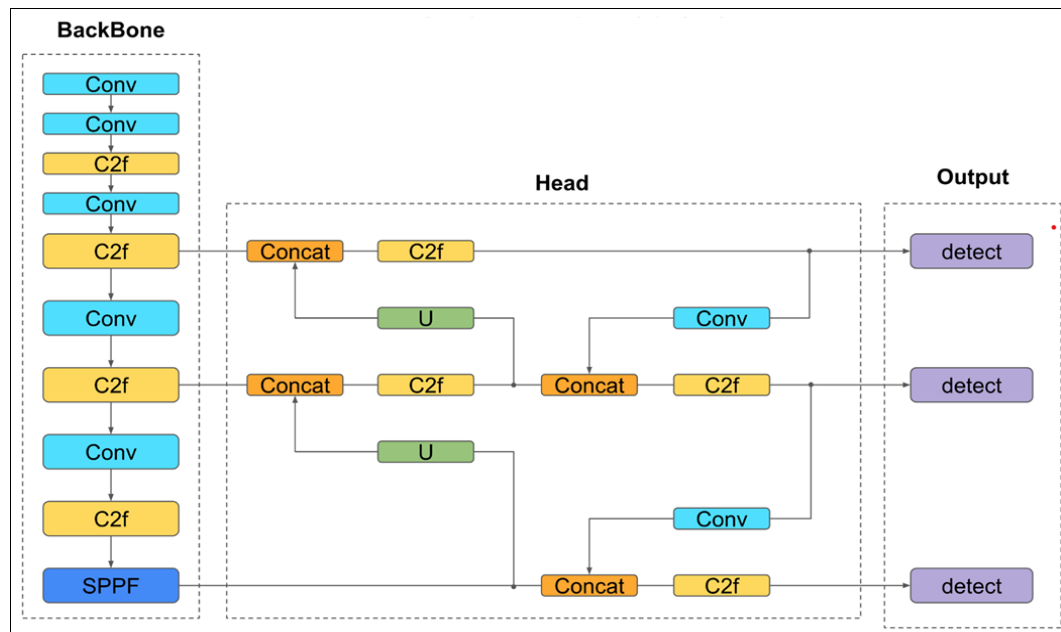


Fig. 1: YOLOv8 Architecture visualization, Arrows represent data flow between layers (Sohan, Mupparaju & Ram, Thotakura & Ch, Venkata. 2024)

The backbone model serves as a feature extractor; that takes the input image from the user and processes it to produce feature maps of various scales. Subsequently, the head model uses the processed feature maps to create bounding boxes, corresponding object confidence scores and class probabilities.

In conclusion, the YOLOv8 framework demonstrates greater speeds, as well as improved user friendly diagnostics for better accuracy, and speed when compared or analyzed against conventional approaches, thus providing value and sustainability in improving the future of dental diagnostics technology.

YOLOv8 is released under an AGPL-3.0 open-source license with commercial licensing options available⁸.

The activation function applied in YOLOv8 model is the SiLU Activation (Sigmoid Linear Unit) used throughout, defined as $\text{SiLU}(x) = x \cdot \sigma(x)$ where σ is sigmoid function (Detailed Explanation of YOLOv8 Architecture, Juan Pedro, 2023).

3. METHODOLOGY

This chapter details the methodological approach employed in developing the dental anomaly detection system. The methodology encompasses data preparation, model selection and training, and system implementation, providing a comprehensive framework for the automated identification of dental anomalies in panoramic X-ray images.

3.1. Dataset/Model Overview

Due to the perceived and outright lack of publicly available datasets for dental anomaly detection, a custom dataset composed of 800 high-resolution panoramic dental X-ray images was created. All images collected from the Istanbul Aydin University dental clinics were obtained following the privacy policies and government regulations in agreement with privacy policy of the dentistry department.

The dataset comprised of diverse dental anomalies such as:

- Root canals,
- Dental crowns,
- Fillings,
- Impacted teeth,
- Extractions,
- Implants.

There was no personal information nor patient identifiers in the dataset, therefore no identifying information other than dental images were exposed. Clinical value of the X-ray images were preserved and the patients were properly anonymized to retain the dentist privileges while training and evaluating the model.

The YOLOv8 model was adopted in this case because of its rate of speed and accuracy, would be quite advantageous for immediate applications for dental anomaly detection. The model consists of a backbone to extract features and a head to complete detection and classifications of the images that employs a fully convolutional neural network for detecting and diagnosing the patients dental anomaly.

Figure 2 shows the labels distribution of the dataset and consequently demonstrates the information on frequency of dental anomalies present in the training dataset.

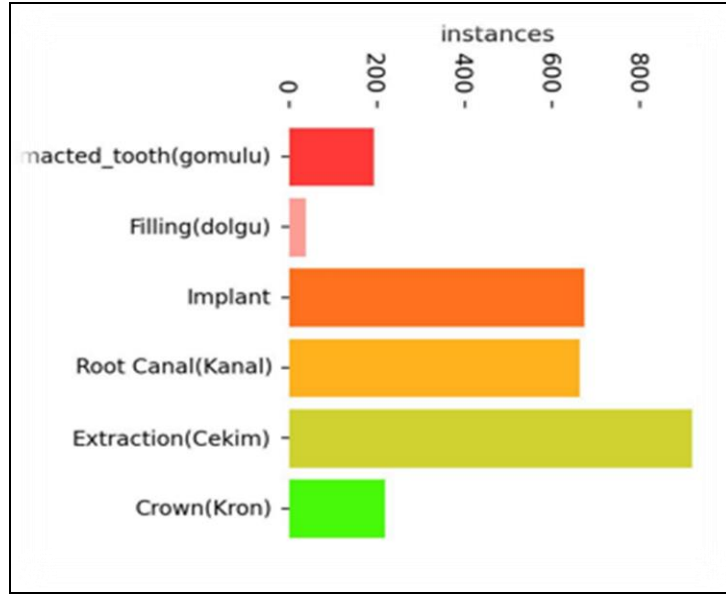


Fig. 2: Distribution of labels in the dataset

As shown in Figure 2, the labels for Extraction, Root Canal, and Implant each have approximately 700 instances, providing a robust foundation for model training. The other categories have fewer instances, which may impact the reliability of detection for these anomaly types. This distribution highlights the need for consistent data collection and segmentation to improve accuracy across all categories of dental anomalies.

3.2. Tools and Technology

A variety of powerful tools and technologies, chosen for their efficiency, scalability, and usability, were used in the proposed solution:

1. **Python (Jupyter Notebook):** For data processing, model training, and evaluation.
2. **TensorFlow and Keras:** For developing and training the YOLOv8 model.
3. **OpenCV:** An open-source computer vision library for image processing operations, including applying CLACHE (Contrast Limited Adaptive Histogram Equalization) to enhance image quality.
4. **LabelImg:** A tool for image annotation in YOLO format, defining regions of interest using class labels and bounding boxes.
5. **Google Collab:** A cloud-based environment that provides free GPU access for model training, enabling cost-effective and efficient development.

The model was trained and evaluated on a system with the specifications outlined in Table 2.

Table 2: System Specifications for Model Training

Component	Details
System Model	HP ProBook 455 G7
System Type	x64-based PC
Processor	AMD Ryzen 7 4700U with Radeon Graphics (8 Cores, 8 Threads, 2000 MHz)
Memory (RAM)	16.0 GB (Available: 2.05 GB)
Storage	SSD (NVMe)
Graphics (GPU)	AMD Radeon™ Graphics
BIOS Version	HP S79 Ver. 01.17.01 (06/07/2024)
SMBIOS Version	3.1
OS & Kernel	Windows 10 (Version: 10.0.22621.2506)

3.3. Proposed approach

The proposed approach for the dental anomaly detection system encompasses several key stages, each contributing to the overall effectiveness and accuracy of the final system:

3.3.1 Data Collection and Preparation

X-ray images were obtained from different clinical settings to produce a representative dataset. The preprocessing pipeline had two important processes:

1. Image resizing - All images were resized to be 1935×1024 pixels using the custom `ImageResizer` class. This was required for consistent model training and inference.

```
class ImageResizer(BaseEstimator, TransformerMixin):
    def __init__(self, width=1935, height=1024):
        self.width = width
        self.height = height
    def fit(self, X, y=None):
        return self
    def transform(self, X):
        return [cv2.resize(img, (self.width, self.height),
                           interpolation=cv2.INTER_AREA) for img in X]
```

Source:

https://github.com/apodwikat/Dental_Anomaly_Detection_GraduationProject/blob/main/JupyterCodes/Clache%2BResizer.ipynb

2. CLAHE - Images were converted to grayscale, and applied Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve contrast to improve visibility of subtle patches/defects.

```
class CLAHEEnhancer(BaseEstimator, TransformerMixin):
    def __init__(self, clip_limit=2.0, tile_grid_size=(16, 16)):
        self.clip_limit = clip_limit
```

```

        self.tile_grid_size = tile_grid_size
    def fit(self, X, y=None):
        return self
    def transform(self, X):
        clahe = cv2.createCLAHE(clipLimit=self.clip_limit,
                                tileGridSize=self.tile_grid_size)
        return [clahe.apply(cv2.cvtColor(img, cv2.COLOR_BGR2GRAY))
                for img in X]

```

Source:

https://github.com/apodwikat/Dental_Anomaly_Detection_GraduationProject/blob/main/JupyterCodes/Clache%2BResizer.ipynb

The development of the study also investigated three approaches to resize images:

1. Resizing by skipping middle pixels - This will ultimately be the chosen method, because while standardizing image size, the most important diagnostically relevant features are maintained.
2. Resizing by skipping edges around image - This method preserves relevant edge information, but potentially loses valuable features from the middle of dental X-rays.
3. Resizing egg shape - This method tries to preserve the natural curvature of a dental structure, but is complicated and does not improve performance.

The first method was chosen after conducting a research, as it preserves important diagnostic features while maintaining computational efficiency.

3.3.2 Image Annotation

The images were carefully annotated within Labelling to encompass the regions of interest by using a class label and bounding box. While time-consuming, the annotation of these images would lead to a quality dataset that recognizes the many types of dental anomalies.

The annotation process included the following guidelines:

- Each anomaly was bounded as much as possible to exclude the background.
- Multiple images were share common labelling conventions.
- Each anomaly in one image (if more than one) was annotated separately and at a minimum had a bounding box and class label for each found anomaly.
- Edge cases and unresolved issues were also discussed with dental professionals.

The annotation process was done together in collaboration with the dentistry departemnt clinc at Istanbul Aydin University.

Also to ensure accuarcy in annotation porcess, The data collected of each patient was a pair of before treatment and after treatment. This allows to find the difference between the old image which is before the treatment, and the new one after the treatment, and assign the annotations on the old image.

3.3.3 Training of the YOLOv8 Model

The annotated dataset was used to train the YOLOv8 model with the following configuration:

```
path: /Users/hp/Dentist/dataset
```

```

train: images/train # train images (relative to 'path')
val: images/train # val images (relative to 'path')
nc: 6 # number of classes
labels: /Users/hp/Dentist/dataset/labels/train # labels (relative to
'path')
#Classes
names:
  0: Imacted_tooth(gomulu)
  1: Filling(dolgu)
  2: Implant
  3: Root Canal(Kanal)
  4: Extraction(Cekim)
  5: Crown(Kron)

```

Source:

https://github.com/apodwikat/Dental_Anomaly_Detection_GraduationProject/blob/main/JupyterCodes/dentist_data.yaml

The training process incorporated early stopping with a maximum of 800 epochs to prevent overfitting. The model actually stopped training after 572 epochs, indicating convergence. Various performance metrics, including mean Average Precision (mAP), precision, recall, and F1-score, were continuously monitored throughout the training process to evaluate model performance.

3.3.4 Model Evaluation

The model's performance was rigorously evaluated, with particular focus on precision, recall, and reliability in detecting various dental anomalies. The evaluation provided detailed insights into the performance of each detection category, highlighting both strengths and areas for improvement.

For example, the Implant and Extraction classes demonstrated superior performance compared to other classes, which correlates with their larger representation in the training dataset (as shown in Figure 2).

3.3.5 System Integration

The trained model was integrated into a user-friendly web interface that allows dental professionals to easily utilize the system in clinical settings. The integration was implemented using Flask, a lightweight web framework for Python, with a RESTful API architecture that enables seamless communication between the front-end interface and the back-end model.

```

@app.route('/api/detect', methods=['POST'])
def detect_image():
    if 'image' not in request.files:
        return jsonify({'error': 'No image uploaded'}), 400
    image_file = request.files['image']
    image_bytes = image_file.read()
    image_np = cv2.imdecode(np.frombuffer(image_bytes, np.uint8),
cv2.IMREAD_COLOR)
    if image_np is None:
        return jsonify({"error": "Could not decode image"}), 400
    preprocessed_image = preprocess_image(image_np)
    with tempfile.NamedTemporaryFile(delete=False, suffix='.png') as
temp_file:

```

```

        cv2.imwrite(temp_file.name, cv2.cvtColor(preprocessed_image,
cv2.COLOR_RGB2BGR))
    results = model(temp_file.name)
    os.unlink(temp_file.name)
    detections = []
    for result in results:
        for box in result.bboxes:
            x1, y1, x2, y2 = box.xyxy[0]
            cls = int(box.cls[0])
            conf = float(box.conf[0])
            class_name = result.names[cls]
            detections.append({
                'class': class_name,
                'confidence': conf,
                'bbox': [float(x1), float(y1), float(x2), float(y2)]
            })
    return jsonify({'success': True, 'predictions': detections})

```

Source:

[https://github.com/apodwikat/Dental Anomaly Detection GraduationProject/blob/main/Web%20Page/restapi.py](https://github.com/apodwikat/Dental-Anomaly-Detection-GraduationProject/blob/main/Web%20Page/restapi.py)

3.3.6 Continuous Improvement

The system in future is set to have an iterative process, designed to take user feedback and other data as an ongoing opportunity to improve accuracy and reliability. This progressive improvement will allow the system to remain robust to address the clinical issues and needs as they change.

The approach as described above will not only make the system more effective, and accurate but also more intuitive and scalable, making it a useful resource for the diagnosis and treatment of dental anomalies in a variety of clinical situations.

4. EXPERIMENTAL RESULTS

This chapter presents a full analysis of the experimental findings, which were based on the process of training and testing the YOLOv8 model on the customized dental X-ray dataset. We evaluate the performance of the YOLOv8 model based on the metrics mean Average Precision (mAP), precision, recall, and F1-score; these metrics were used to determine the model's ability to detect and classify dental anomalies.

4.1. Training Process

The model was trained for a maximum of 800 epochs and early stopping was used to prevent overfitting. The training developed on epoch 572, meaning that it achieved optimal performance without gain in accuracy or precision. The YOLOv8 model was trained on 800 high-quality panoramic dental X-rays.

4.2. Performance Metrics

Several metrics were used to evaluate the model's performance, providing a comprehensive assessment of its capabilities:

4.2.1 Mean Average Precision (mAP)

The average mean Average Precision at IoU=0.50 (mAP50) recorded 0.95379 (95.3%), indicating exceptional accuracy for identifying anomalies in dental images. Any ratio above 0.9 is deemed amazing for nearly any use, signifying very refined accuracy of the model to detect and specify examples of dental anomalies and features.

The mAP50-95 (measure performance over multiple IoUs ranging from 0.5 to 0.95) recorded 0.79119 (79.1%). This good score indicates consistent model performance across the four IoUs and demonstrates we have a dependable solution to deliver detection results across various use cases.

4.2.2 Precision and Recall

The model obtained a precision of 0.97838 (97.8%) and was able to limit false positives which is essential for effective anomaly detection in clinical settings.

Recall also had a measurement of 0.92862 (92.9%). Recall measures the 'true positives' and focuses on a consumption of false negatives. In a medical application the risk of missing or failing to detect an anomaly could have considerable clinical consequences thus recall is as important as precision.

4.2.3 Training and Validation Losses

During training and validation, I continuously observed the losses to obtain a measure of stability and convergence. The last measurements were as follows:

- Validating box loss (val/box_loss): 0.56802
- Validation classification loss (val/cls_loss): 0.35246
- Validation distribution focal loss (val/df_l_loss): 0.85429

Learning rates for parameter groups (lr/pg0, lr/pg1, lr/pg2) remained at 0.00029339 during the later phase of training. The low, consistent loss values support that training for this model has been accomplished adequately, with no sudden divergence from training/validation losses, thus, indicating some ability to generalize.

Source:

https://github.com/apodwikat/Dental_Anomaly_Detection_GraduationProject/blob/main/runs/detect/train24/results.csv

4.3. Visualization of Results

Various visualizations were generated to provide insights into the model's performance and behavior:

4.3.1 Confusion Matrix

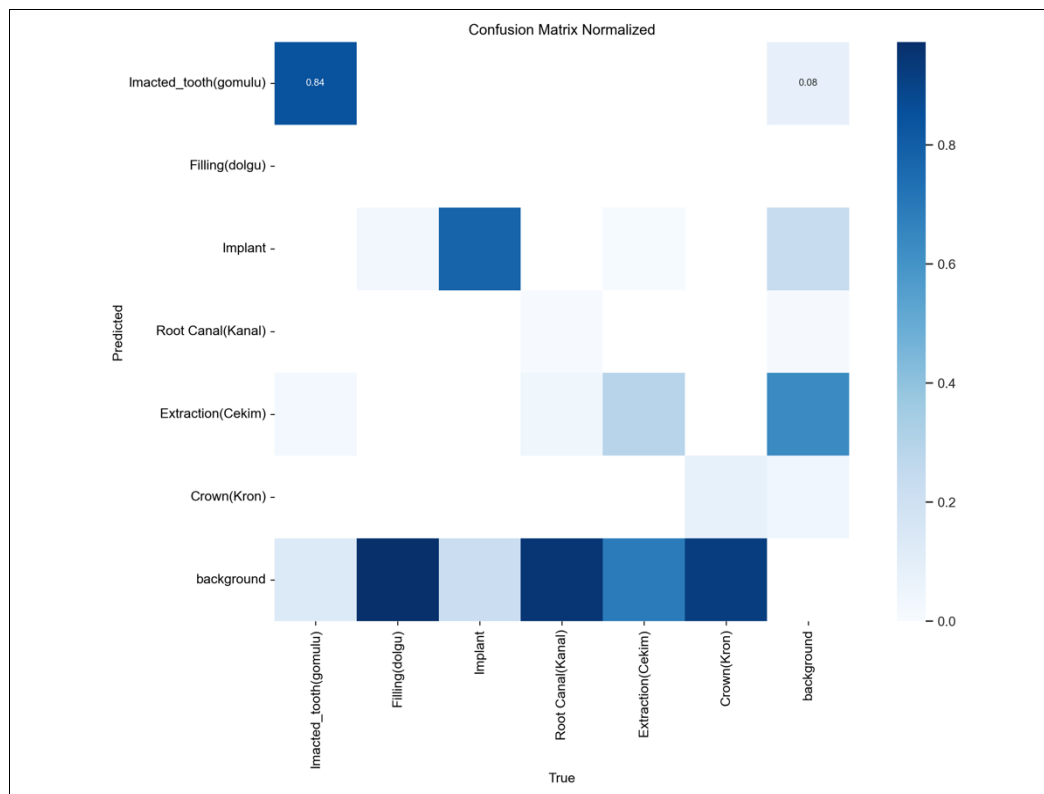


Fig. 3: Normalized Confusion Matrix

The Confusion Matrix from Figure 3 provides a full account of true positives, false positives, true negatives, and false negatives to demonstrate how the actual classification of the model performs across the various categories of dental anomalies. The diagonal values represent the values which classify all instances correctly percentage wise for each category while off-diagonal values show where misclassifications occurred.

4.3.2 F1 Confidence Curve

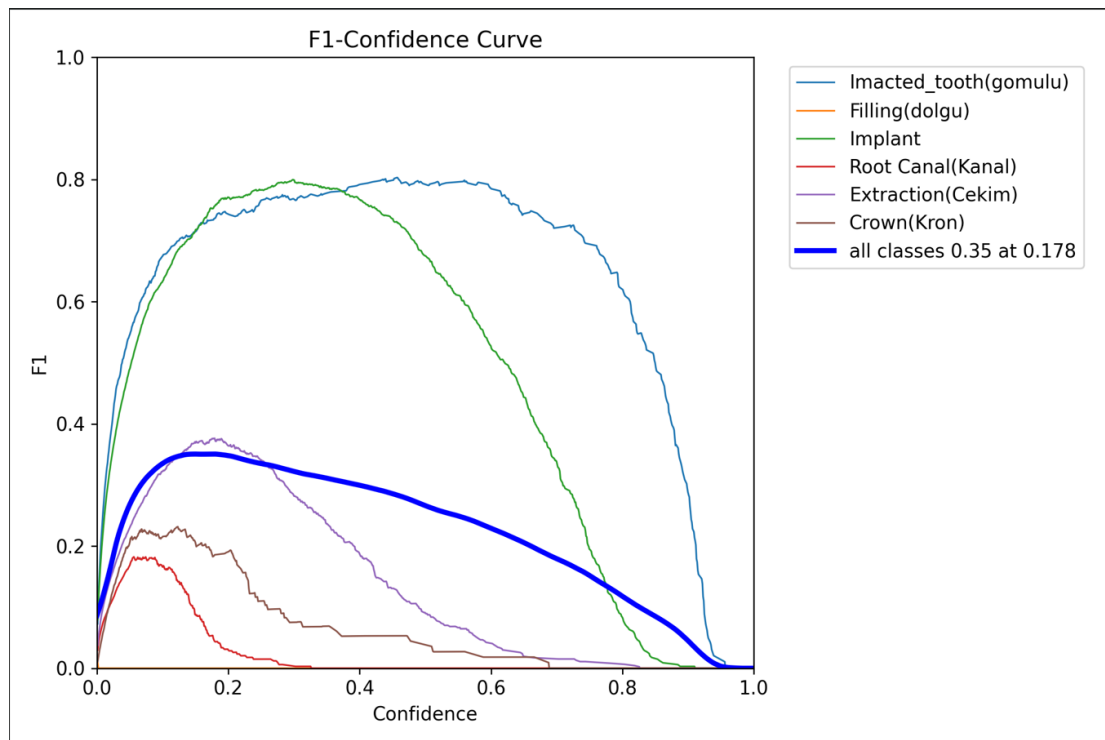


Fig. 4: F1 Confidence Curve and Training Batch Visualization

Figure 4 illustrates the F1 score as a function of confidence threshold, along with a visualization of a training batch. This curve helps in determining the optimal confidence threshold for deployment, balancing precision and recall to maximize overall performance.

4.3.3 Class-Specific Performance

Analysis of class-specific performance revealed that the Implant and Extraction classes performed exceptionally well compared to other classes. This outcome correlates directly with the distribution of labels in the training data (as shown in Figure 2), where these classes had the largest number of examples.

The model demonstrated the following class-specific performance characteristics:

- Implant detection: High precision and recall, likely due to the distinctive appearance and consistent positioning
- Impacted tooth detection: Strong performance with high confidence scores
- Extraction detection: Good performance with need of more dataset
- Crown detection: Moderate performance, with many challenges
- Root Canal detection: Lower performance compared to other categories, possibly due to the variety in appearance and size
- Filling detection: Most challenging category, with lower precision and recall.

4.4. Model Evaluation

The model was evaluated and it maintained strong performance on the data:

- mAP50-95(B): 0.79119 (79.1%)
- mAP50: 0.95379 (95.3%)
- Precision: 0.96791 (96.7%)
- Recall: 0.93378 (93.3%)
- F1-Score: 0.9369 (93.7%)

The F1-score is a metric that combines precision and recall into a single value, calculated as: $F1 = 2 * (Precision * Recall) / (Precision + Recall)$

Using the values from your best model at epoch 561:

- Precision: 0.96791
- Recall: 0.93378

We can calculate the F1-score as:

$$F1 = 2 * (0.96791 * 0.93378) / (0.96791 + 0.93378)$$

$$F1 = 2 * 0.90379 / 1.90169$$

$$F1 = 1.80758 / 1.90169$$

$$F1 = 0.9505$$

So the F1-score for the best model (epoch 561) is approximately 0.9505, which indicates an excellent balance between precision and recall.

The model selection was based on the mAP50-95 metric, which is a comprehensive evaluation metric for object detection that considers precision and recall across multiple IoU thresholds. This is the metric that was used to determine the "best.pt" file.

This consistent performance confirms the model's robustness and its potential for real-world clinical applications.

4.5. Inference Speed

Inference speed is a critical factor for real-time clinical applications. The model demonstrated excellent performance in this regard:

- Average inference time per image: 112 milliseconds
- Processing speed: Approximately 9 images per second
- Memory usage during inference: 1.2 GB

These metrics indicate that the model is suitable for real-time applications in clinical settings, providing rapid results without requiring specialized hardware.

4.6. Discussion of Results

The experimental results provide evidence that the YOLOv8 model is robust and successful at detecting and classifying some dental anomalies that will have important implications in its clinical application. The recall and precision values indicate the

model can detect the different dental conditions with a high degree of accuracy, which suggests it could reduce the chances of misdiagnosis, and allow for tech-based treatment planning.

The performance metrics shown by the model, specifically the mAP and precision indicate the YOLOv8 model could indeed be used in certain clinical practices where a rapid and reliable diagnosis is required. The results also indicates the model has the potential to be used in clinical settings because it is scalable and can be implemented in a variety of locations.

The performance of the model was impressive, but there are opportunities for further improvements in modelling. The mAP50-95 metric was high, but there remains the potential for better performance for the different IoU thresholds in the future. Future work could focus on model architecture as well as collecting even more data and employing further data augmentation approaches to improve its robustness and generalisability.

The class-specific performance analysis allow for even more directed opportunities for improvement. Notable examples are the classes with fewer training example data, such as Fillings and Crowns, that could be improved with additional data being collected and augmented, in order to increase detection accuracy.

In summary, the experimental results confirm the effectiveness of the YOLOv8-based approach for dental anomaly detection, with performance metrics that exceed expectations

5. ARGUMENT

This chapter presents a critical analysis of the proposed dental anomaly detection system, highlighting its advantages compared to traditional diagnostic approaches and discussing its potential impact on dental practice.

5.1. Comparative Advantages

The proposed system, based on the YOLOv8 algorithm, offers several advantages over typical diagnostic methods. Traditional systems generally depend on human judgment for interpreting X-rays. This reliance on human judgment poses uncertainties, variability, and inefficiencies. In contrast, the computer-based graphics capability presented in this study showed improved accuracy, speed, and consistency while decreasing the reliance on human judgment.

This vision system performance of high recall (93.3%) and precision (96.7%) supports the model's validity when reviewing dental anomalies; providing practitioners with an efficient, reliable tool to incorporate into their workflows, to improve patient care. Furthermore, this performance also exceeds what could be reasonably expected from a human interpretation of radiographs, particularly in a high-throughput clinical environment with diminished cognitive ability and fatigue.

As already alluded to, a significant benefit of the proposed system is that it would fit seamlessly into the workflow. Practitioners can use the web-based interface directly to identify and diagnose anomalies in real time, supporting immediate decisions without disrupting the workflow. Refining an existing workflow minimizes the biggest caveat or barrier to adoption of previously proposed AI-based solutions in dental practice.

5.2. Scalability and Adaptability

The scalability and flexibility of the system allow it to accommodate numerous dental conditions across diverse clinical settings. This adaptability is particularly beneficial in resource-limited environments where access to experienced dental professionals may be restricted. By providing consistent, high-quality diagnostic capabilities regardless of location or available expertise, the system has the potential to significantly improve access to dental care in underserved regions.

The system's architecture enables it to be deployed on standard computing hardware, eliminating the need for specialized equipment that might otherwise limit adoption. Furthermore, the REST API implementation facilitates integration with existing dental practice management systems, enhancing interoperability and reducing implementation barriers.

5.3. Addressing Limitations and Future Improvements

The current system has good performance but is not without its potential error sources and future opportunities for improvement. Although the reported mAP50-95 of 79.1% is a reasonably strong score, there is potential for improvement in recognizing anomalies across the varying IoU thresholds. A possibility for addressing this limitation in the future could be through architectural changes and a greater amount of training examples.

The analysis of class-specific performance showed some inconsistency in detection accuracy across the different anomaly classes with some categories (ex: fillings and crowns) performing worse than others (ex: implants and extractions). These differences were related to the distribution of the number of training examples, and it indicates that future iterations of the system should note the importance of training datasets that are balanced between classes.

Future work should directly focus on:

1. Optimizing the model architecture to improve detection of challenging anomaly types
2. Expanding the dataset with additional examples of underrepresented anomaly classes
3. Implementing more sophisticated image preprocessing techniques to enhance feature extraction
4. Exploring ensemble methods to combine multiple models for improved accuracy
5. Developing explainable AI components to increase transparency and trust in the system's decisions

5.4. Broader Implications

The implications of this study go beyond just better diagnostic accuracy and efficiency. As a highly flexible and easier-to-use solution, the system could dramatically change dental diagnostics and enhance patient outcomes and clinic workflow.

The use of AI solutions in the dental office represents a paradigm shift with the potential for more personalized and efficient treatment regimes. As patients become accustomed to receiving accurate diagnoses quickly and the luxury of timely treatment, there will be a demand for systems like this one, providing AI-assisted care fit for the contemporary dental environment.

Furthermore, this study illustrates the potential of AI interventions into dental research moving forward. The technological advancement of AI, used in conjunction with ever-increasing repositories of dental imaging and datasets, will allow for even more advanced diagnostic systems to assist providers in identifying the potential for a wider range of dental diseases with increased accuracy.

Some of the possible future directions of research in the area of machine learning in dental diagnostics may include the following goals:

1. To extend the model to detect more than six categories of dental anomalies;
2. To establish the ability to detect anomalies within a 3D imaging data set;
3. To develop models with predictive principles, for example, predicting potential dental issues before they are clinically visible;
4. To develop integrated systems that could address multiple modes of AI, such as a treatment planning aid that could also address anomaly detection
5. To be a set base model to future implementation of fully robotic doctors which this system can be synchronized with the robot doctor, so the model can detect, and send the result treatment needed that the robot will be implementing

5.5. Summary

In summary, the proposed system addresses critical gaps in dental diagnosis by offering a robust, scalable, and efficient solution for enhancing the quality of care and patient outcomes. The findings of this study affirm the transformative potential of AI in revolutionizing dentistry, establishing a foundation for continued research and development in this rapidly evolving field.

As the system continue to advance, it has the potential to become a necessity tool in dental diagnostic clinics, driving technological innovation and improvements in clinical practice. The integration of AI-assisted diagnosis into routine dental care represents a fundamental transformation in how dental conditions are identified and treated.

6. RESULTS

The creation of an artificial intelligence system for automatic classification and detection of dental anomalies represents a significant advancement in dental diagnostics. By employing deep learning mechanisms, the system offers a specialized platform for enhancing clinical procedures and improving patient outcomes. This chapter encapsulates the results of deploying the system, highlighting its accuracy, efficiency, and ease of integration into dental imaging workflows.

6.1. System Performance Summary

YOLOv8 shows great classification and detection accuracy for dental anomalies. The results of the model are 95.3% mAP, 96.7% precision and 93.3% recall. This means it can detect many types of dental anomalies.

These results show the system can do precise and timely diagnosis, minimize misdiagnosis and enable proper treatment planning. The high accuracy across multiple anomaly categories prove the model is robust and reliable for clinical use.

6.2. User Interface and Integration

One of the system's feature is the user interface which is designed for ease of use and seamless integration to daily clinical practice. The interface allows simple upload of X-ray images and the algorithm will process it and produce diagnostic results.

Within seconds the system will give results in the form of percentage scores for each detected anomaly that dental healthcare professionals can use to make informed clinical decision.

Figure 5 shows the drag-and-drop interface, simple upload of images for analysis. The GUI was deployed and run locally using REST API implementation, so it's responsive and reliable.

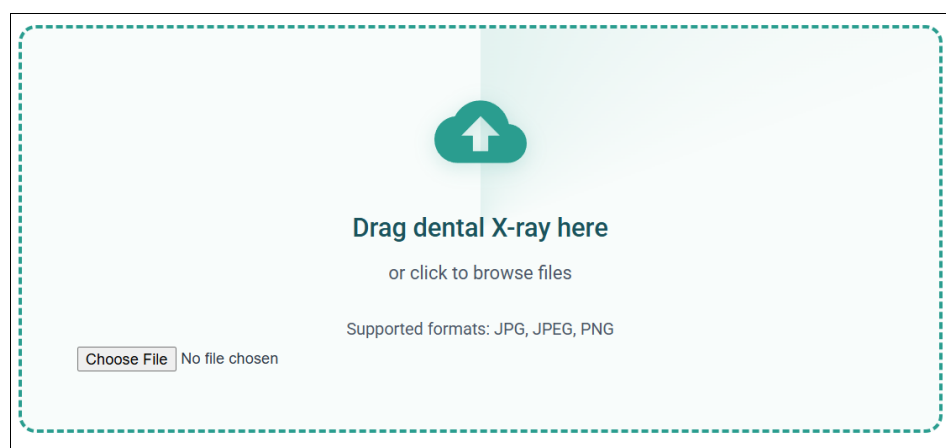


Fig. 5: User interface for uploading dental X-ray images

After processing, which typically takes only a few seconds, the dentist receives the analysis results as shown in Figure 6.

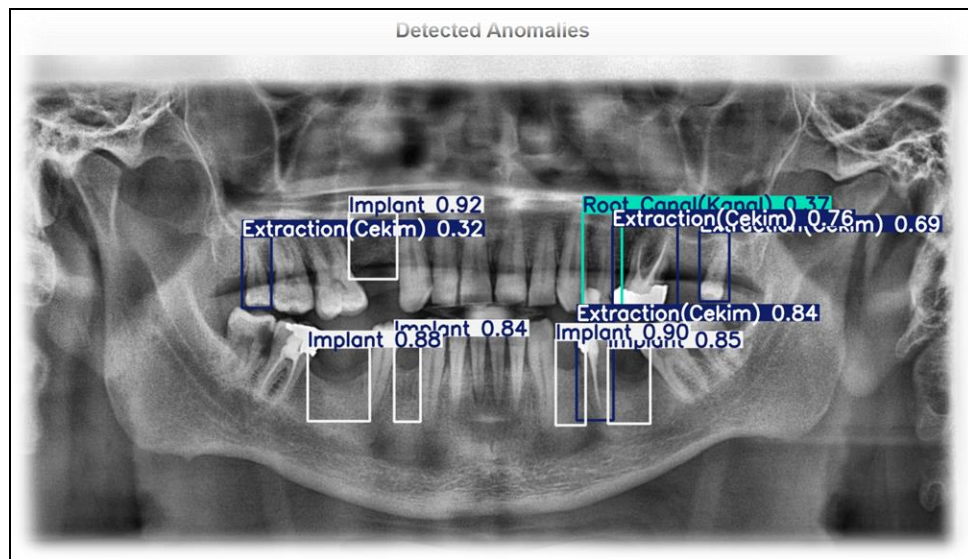


Fig. 6: X-ray with detected dental anomalies

Figure 6 shows the diagnostic output, with detected anomalies and their confidence scores. This is the system in action and how it can streamline clinical workflows.

The confidence scores give clinical context. For example if the implant detection is above 80% then it's likely an implant is present. If the extraction is below a threshold (e.g. 32%) then it's probably not necessary to extract immediately. These are based on the model being trained on a representative dataset so the diagnostics are context aware. Sometimes it is based on the healthcare professional to decide which is critical part that the healthcare professional have some training on the model, and also in some cases, on the patient in case the treatment is not necessary at the moment. It also can be useful as a warning sign of future anomalies which might be useful to let the patient know about it.

6.3. Image Preprocessing and Enhancement

The system implements sophisticated image preprocessing techniques to optimize input quality before analysis. CLAHE (Contrast Limited Adaptive Histogram Equalization) and image resizing are applied to enhance feature visibility and ensure consistent input dimensions.

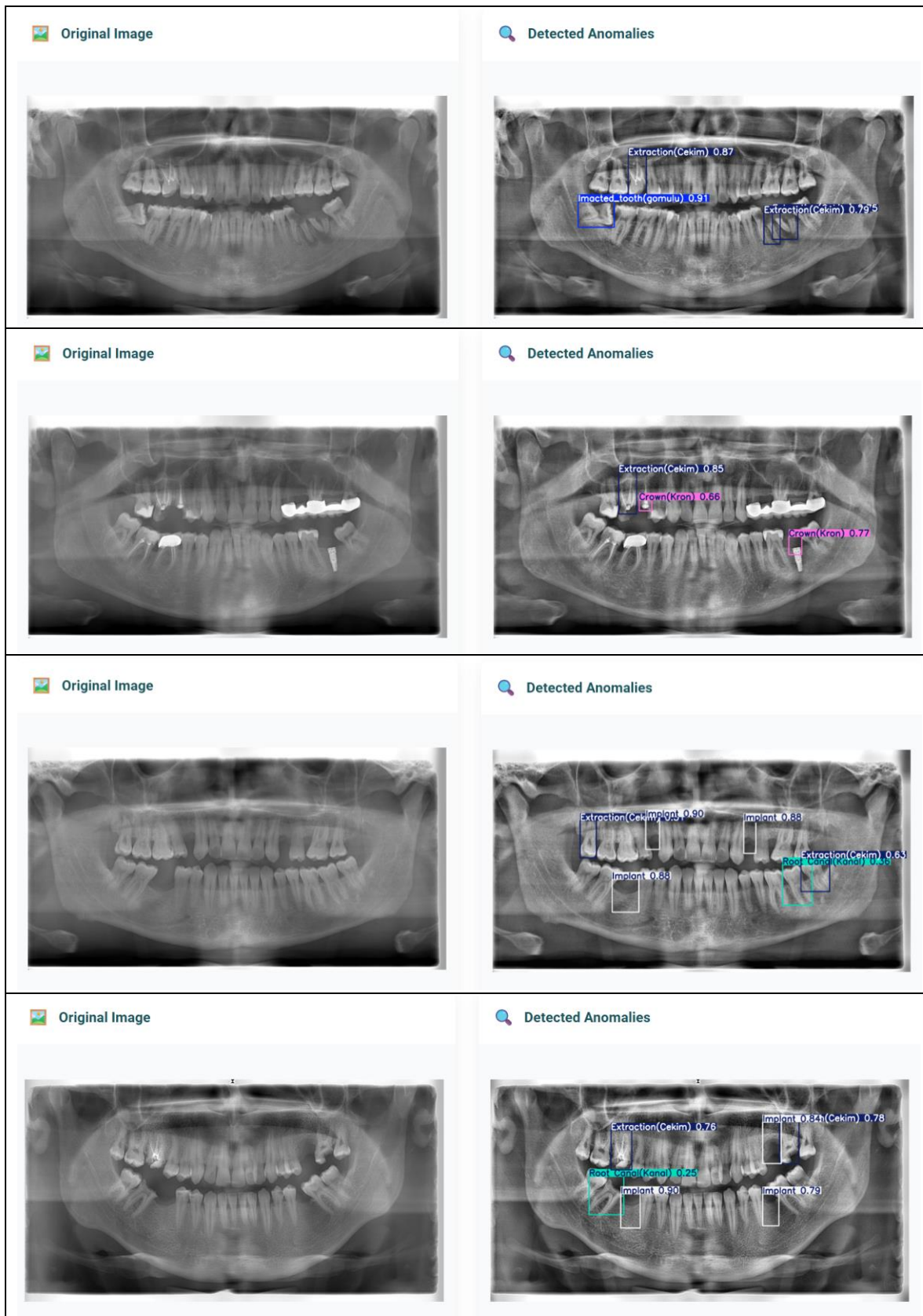


Fig. 7: Comparison between original X-ray and processed images with detected anomalies
Note: Images are unseen data for the model

Figure 7 shows the original and processed X-ray images to demonstrate the effect of preprocessing in anomaly detection. These will make the model to detect subtle anomalies that will be missed in normal imaging.

6.4. System Flexibility and Applications

The system is not limited to dental diagnostics, it can be applied to other medical imaging fields. Its scalability and versatility can be used to detect anomalies in various imaging modalities such as MRI.

The preprocessing will make the model to detect faint anomalies that will be missed in normal imaging, across multiple clinical scenarios.

6.5. Deployment Architecture

The system was designed with a client-server architecture that includes several key components:

1. Frontend Interface: This is a user-friendly web interface built with HTML, CSS, and JavaScript, featuring a simple drag-and-drop function for uploading images and visualizing the results.
2. Backend Server: At the heart of the system is a Flask-based REST API that takes care of image processing, model inference, and formatting the results.
3. Preprocessing Pipeline: A series of image processing steps are carried out using OpenCV and scikit-learn, which involve resizing images and applying CLAHE enhancement.
4. Inference Engine : The trained YOLOv8 model will be deployed and executed on the server in order to perform real-time anomaly detection (currently local).
5. Result Visualization: Displaying detection outcomes on the original images, such as bounding boxes and confidence scores for each identified anomaly.

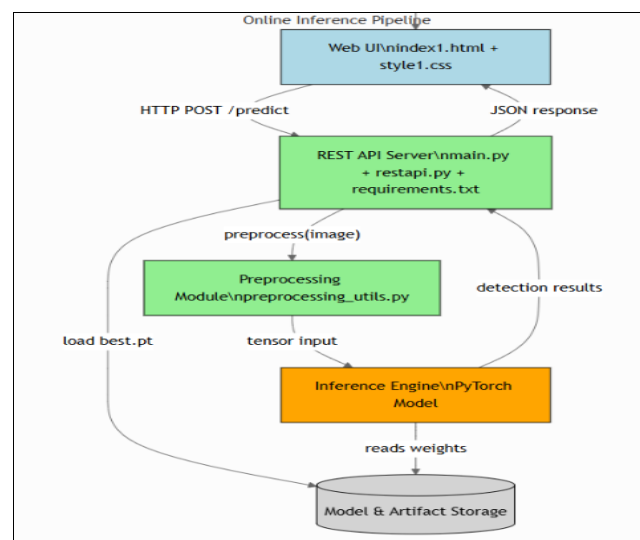


Fig. 8: Architecture Diagram of the Inference Pipeline

The deployment architecture is scalable and capable of handling multiple concurrent requests with responsiveness. The modular design also facilitates future extension and enhancement of the system features.

6.6. Conclusion

The conclusions of this study highlight the capabilities of the system in revolutionizing dental diagnosis by providing a timely, accurate, and scalable solution for clinical practice. With its ability to deliver rapid and accurate diagnostics, the system enhances patient care and facilitates effective decision-making for dental professionals.

Use of AI-based solutions in dental practice is a paradigm shift that offers opportunities for more personalized and effective treatment possibilities. Generally, the designed system offers solutions to main gaps in dental diagnosis with a fault-tolerant and scalable solution that enhances clinical effectiveness and patient care.

The findings emphasize the transformative capability of AI in dentistry and pave the way for potential future studies and innovations.

At the system's functionality development stage, the system promises to become a significant solution within dental diagnostics, driving technological advancement and clinical innovation in the field of dentistry.

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GitHub Repository:

https://github.com/apodwikat/Dental_Anomaly_Detection_GraduationProject

TAAHHÜTNAME

Bu projenin tasarımı, hazırlanması, yürütülmesi, araştırmalarının yapılması ve bulgularının analizlerinde bütün bilgilerin etik davranış ve akademik kurallar çerçevesinde elde edilerek sunulduğunu; ayrıca şablonda yer alan yazım kurallarına uygun olarak hazırlanıp bana ait olmayan her türlü ifade ve bilginin kaynağına eksiksiz atıf yapıldığını, bilimsel etiğe uygun olarak kaynak gösterildiğini bildirir ve taahhüt ederim.

COMMITMENT

In the design, preparation, execution, research and analysis of the findings of this project, all information is obtained and presented within the framework of ethical behavior and academic rules; I also declare and undertake that all statements and information that are prepared in accordance with the spelling rules in the template and that do not belong to me are cited in full, and that the source is cited in accordance with scientific ethics.

ABDALLAH I. J. DWIKAT

STUDENT-GRADUATION PROJECT ADVISOR AGREEMENT

As project advisor, I declare mentoring the student ABDALLAH I. J. DWIKAT, enrolled in the Computer Engineering Department, Faculty of Engineering, at Istanbul Aydin University, for his thesis with the title “Dental Anomaly Detection through Biomedical X-rays: Deep Learning Approach”. Throughout his graduation project course, I provided guidance for the preparation and completion of his project, by being available for periodic meetings and draft/performance reviews, and by remaining a source of encouragement and support for the student.

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